Empirical Case Study: The "Breaking Bad" Worldview Divergence Model

To illustrate how the math used in theorems operate in practice, we examine a concrete simulation mapping the moral divergence of **Hank Schrader** (the rule-following lawman) and **Walter White** (the chemist who turned into the drug lord "Heisenberg") across different life contexts.

Setup & Parameterization

- **Moral State Space ($\mathbb{R}^3$):** Valuations are projected onto three axes: Order/Safety (x_1), Fairness/Justice (x_2), and Autonomy/Self-Preservation (x_3).
- The Lie group $G = \mathrm{SO}(3)$ represents rotations of these emotional axes.
- **Rotation notation**: $R_{i\to j}(\theta)$ is the identity matrix except for entries $(i,i) = (j,j) = \cos\theta$, $(i,j) = -\sin\theta$, and $(j,i) = \sin\theta$. It moves weight from axis i toward axis j.
- **Worldview of an agent**: Agent k has a loop of experiences with holonomy $H_k \in SO(N)$, which is the product of the edge rotations along the loop. Its routine target is $t_k = H_k u_0$, representing the shared baseline u_0 carried around its own loop.
- **Context/Places:** Home, Street, Doctor, School/Work which can effect on the worldview
- **Context Loops:**
    - **School/Work Loop:** Home → School/Work → Home (shared institutional structure)
    - **Street Loop:** Home → Street → Home (unshared/high-risk environment)
    - **Doctor Loop:** Home → Doctor → Home (cancer diagnosis)

Our three moral/emotional valuation axes are:

1. **Order/Safety (x_1):** Institutional law, safety, and conventional order.
2. **Fairness/Justice (x_2):** Procedural integrity, institutional fairness, and procedural ethics.
3. **Autonomy/ Self-Preservation (x):** Personal sovereignty, dominance, and self-preservation at all costs.

In the beginning, both Hank and Walt share a conventional, law-abiding baseline valuation vector. They place weight on Order/Safety and Fairness, with near-zero valuation placed on autonomous, law-breaking power. The normalized unit vector for high order/safety and basic fairness/justice is:

$$v_0 = \frac{1}{\sqrt{0.95^2 + 0.3^2 + 0.05^2}} \begin{pmatrix} 0.95 \\ 0.30 \\ 0.05 \end{pmatrix} = \frac{1}{\sqrt{0.995}} \begin{pmatrix} 0.952 \\ 0.301 \\ 0.050 \end{pmatrix}$$

The Rule-Follower (Hank):
The loops of interest are Home → Street → Home. Hank's trips onto the street as a police officer and DEA agent rotate his frame slightly and return exactly, whereas Walt's trips rotate it strongly and return with a residual twist. Hank's transition matrices $U_{\text{Hank}}(e)$ are nearly symmetric and **path-independent symmetry**. His holonomy (holonomy measures how a valuation changes after a loop of contexts) on the street loop remains small, showing high alignment with the suburban safety baseline. Hank's moral framework is anchored to institutional safety and procedure. His rotations are small, constrained, and symmetric (going out to the street and returning home brings his frame back very close to baseline).

Home → Street edge: a minor rotation into the Autonomy/Power axis as he faces danger but tightly bound by procedure. It is a $\theta_{\text{H}} = 15°$ rotation from Safety toward Autonomy ($x_1 - x_3$ plane), and his return edge is its transpose:

$$U_{\text{Hank}}(e_{\text{out}}) = R_{1\to3}(15°) = \begin{pmatrix} \cos(\theta_H) & 0 & -\sin(\theta_H) \\ 0 & 1 & 0 \\ \sin(\theta_H) & 0 & \cos(\theta_H) \end{pmatrix} = \begin{pmatrix} 0.966 & 0 & -0.259 \\ 0 & 1 & 0 \\ 0.259 & 0 & 0.966 \end{pmatrix}$$

Street → Home edge: return path undoes the rotation, because it is exactly the transpose of the outbound rotation:

$$U_{\text{Hank}}(e_{\text{back}}) = U_{\text{Hank}}(e_{\text{out}})^T$$

Multiplying Hank's transition matrices around the closed loop yields the identity matrix. Hank's word view remains unchanged.

$$H_{\text{Hank}} = U_{\text{Hank}}(e_{\text{back}})\, U_{\text{Hank}}(e_{\text{out}}) = I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The Rebellious Agent (Walt): Walt's transition matrices exhibit strong path-dependency and asymmetric updates. Each trip to the doctor or onto the street applies a powerful rotational push toward autonomy and power. Walt's transition matrices exhibit **path-dependent asymmetry**. When Walt goes to the street, his internal frame undergoes a strong rotation into the Autonomy/Self-Preservation vs. Safety plane (the Heisenberg pivot). When he returns home, he does not fully revert; his baseline coordinate frame has shifted slightly.

Home → Doctor edge: A rotation shifting away from Institutional Fairness (e_2) toward Autonomy (e_3). Walt thinks that life is cruel and institutions are not fair after his shock cancer diagnosis, because he cannot afford a full treatment. This can be described by a rotation of $\phi_W = 60°$ around the first axis away from Institutional Fairness (e_2) toward Autonomy (e_3).

$$U_{\text{Walt}}(e_{\text{out}}) = R_{2\to3}(60°) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi_W) & -\sin(\phi_W) \\ 0 & \sin(\phi_W) & \cos(\phi_W) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.500 & -0.866 \\ 0 & 0.866 & 0.500 \end{pmatrix}$$

Doctor → Home edge: the return path is not symmetrical. A diagnosis cannot be undone, leaves a full mark and the visit to the doctor as devastating consequences. Only a small part returns to normal

$$U_{\text{Walt}}(e_{\text{back}}) = R_{2\to3}(-10°) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.984 & 0.174 \\ 0 & -0.174 & 0.984 \end{pmatrix}$$

Rotations in SO(3) are commutative only when they happen around the same axis or if they are very small. Very small angles or infinitesimal are approximately commute, but larger angles are not.

$$H_{\text{Walt}} = U_{\text{Walt}}(e_{\text{back}}) \cdot U_{\text{Walt}}(e_{\text{out}}) = R_{2\to3}(-10°)\, R_{2\to3}(60°) = R_{2\to3}(50°) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.643 & -0.765 \\ 0 & 0.765 & 0.643 \end{pmatrix}$$

Home → Street edge: A rotation shifting away from Order/Safety (e_1) toward Autonomy (e_3). Walt no longer thinks that rule breaking is bad. His reasoning might be "if the world is bad and cruel to me, then it does not matter if I am good or bad. I just want to survive. If I can survive by making drugs, then this is legitimate. It is bad but the world is bad too". Walt's outbound edge is a $\phi_W =$

60° rotation.

$$U_{\text{Walt}}(e_{\text{out}}) = R_{1\to3}(60°) = \begin{pmatrix} \cos(\phi_W) & 0 & \sin(\phi_W) \\ 0 & 1 & 0 \\ -\sin(\phi_W) & 0 & \cos(\phi_W) \end{pmatrix} = \begin{pmatrix} 0.500 & 0 & 0.866 \\ 0 & 1 & 0 \\ -0.866 & 0 & 0.500 \end{pmatrix}$$

Street → Home edge: the return path is not completely symmetrical. Because the place left its mark, his return matrix $U_{\text{Walt}}(e_{\text{back}})$ is not just the transpose. It carries residual deformation from his street experiences, causing a shift toward power by rotating around the middle axis (e_2) away from Safety (e_1) toward Autonomy (e_3):

$$U_{\text{Walt}}(e_{\text{back}}) = R_{1\to3}(-59°)$$

$$H_{\text{Walt}} = U_{\text{Walt}}(e_{\text{back}}) \cdot U_{\text{Walt}}(e_{\text{out}}) = R_{1\to3}(-59°)\, R_{1\to3}(60°) = R_{1\to3}(1°) = \begin{pmatrix} 1 & 0 & -0.017 \\ 0 & 1 & 0 \\ 0.017 & 0 & 1 \end{pmatrix}$$

Calculating the Loop Holonomies (*H*)

The total loop holonomy H for a complete cycle (Home → Street → Home) is calculated by multiplying the sequential transition operators:

$$H_{\text{Hank}} = U_{\text{Hank}}(e_{\text{back}}) \cdot U_{\text{Hank}}(e_{\text{out}}) = I$$

Home → Doctor → Home loop for Walt:

$$H_{\text{Walt}} = U_{\text{Walt}}(e_{\text{back}}) \cdot U_{\text{Walt}}(e_{\text{out}}) = R_{2\to3}(-10°)\, R_{2\to3}(60°) = R_{2\to3}(50°) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.643 & -0.765 \\ 0 & 0.765 & 0.643 \end{pmatrix}$$

Home → Street → Home loop for Walt:

$$H_{\text{Walt}} = U_{\text{Walt}}(e_{\text{back}}) \cdot U_{\text{Walt}}(e_{\text{out}}) = R_{1\to3}(-40°)\, R_{1\to3}(60°) = R_{1\to3}(20°) = \begin{pmatrix} 1 & 0 & -0.017 \\ 0 & 1 & 0 \\ 0.017 & 0 & 1 \end{pmatrix}$$

What the Resulting Matrices Show:

1. **Hank's Holonomy (H_{Hank}):** is exactly the identity matrix I_3. His moral coordinate axes are preserved; a trip to a crime scene temporarily stresses him, but his framing snaps back to the suburban safety baseline. After a street trip, his frame returns to the baseline, so his routine target is the baseline itself: $(0.952, 0.301, 0.050)$

2. **Walt's Holonomy (H_{Walt}):** Substantially deviates from the identity matrix I_3.

We can simulating the "Breaking Bad" story by two doctor diagnoses, each followed by street trips, 7 trips in total.

Home → **Doctor** →
Home → Street →
Home → Street →
Home → **Doctor** →
Home → Street →
Home → Street →
Home → Street

Because Hank and Walt experience different context loops (Hank stays anchored to the DEA/suburban loop, while Walt repeatedly traverses the unshared street/meth-trade loop with its dominant attractor their final vectors diverge dramatically. The states of valuation vectors after these 7 loops:

$$v_{Hank} = \begin{pmatrix} 0.952 \\ 0.301 \\ 0.050 \end{pmatrix} \; v_{Walt} = \begin{pmatrix} 0.925 \\ -0.127 \\ 0.358 \end{pmatrix}$$

Hank Schrader's (The Rule-Follower) transition operators stay close to the identity baseline. Under repeated loops, his vector experiences minor perturbations when facing danger, but the spectral pull of his institutional environment draws him right back. Hank remains firmly anchored to institutional rules and safety. His core worldview structure is preserved. Walter White / Heisenberg (The Rebellious Agent) undergoes repeated, asymmetric traversals of the street loop. Walt's valuation vector has essentially inverted. **Safety and Fairness have collapsed**, while **Autonomy/Power (e_3) has completely taken over**. He has moved towards "Heisenberg" vector.

The states of valuation vectors after these 7 loops without doctor diagnosis (only 7 street visits)

$$v_{Hank} = \begin{pmatrix} 0.952 \\ 0.301 \\ 0.050 \end{pmatrix} \; v_{Walt} = \begin{pmatrix} 0.939 \\ 0.301 \\ 0.166 \end{pmatrix}$$

Calculating the Distance
If we measure the distance between Hank and Walt's vectors at the end of the series using our metric space:

1. **At $t = 0$:**

$$\| v_0^{\text{Hank}} - v_0^{\text{Walt}} \|_2 = 0 \quad \text{(they start from the same suburban baseline)}$$

2. **At $t \to 7$:** for target vectors:

$$\left\| \begin{pmatrix} 0.952 \\ 0.301 \\ 0.050 \end{pmatrix} - \begin{pmatrix} 0.925 \\ -0.127 \\ 0.358 \end{pmatrix} \right\|_2 = \sqrt{(0.027)^2 + (0.428)^2 + (-0.308)^2} \approx 0.528$$

Empirical Case Study: Haidt's Moral Foundation & Polarization Model

Summary: Extending the valuation space to Haidt's 6 dimensions (Care, Fairness, Loyalty, Authority, Sanctity, Liberty).

Setup: We have $N = 6$ **dimensions** using Jonathan Haidt's Moral Foundations Theory, and explicitly write out the matrices, vectors, and holonomy mechanics.

The $N = 6$ Moral Valuation Space
An agent's mental state is a unit vector $v \in \mathbb{R}^6$ distributed across Haidt's six foundational axes:

1. e_1: **Care / Harm** (protecting and caring for others)
2. e_2: **Fairness / Cheating** (proportionality, rights, and justice)
3. e_3: **Loyalty / Betrayal** (allegiance to group, patriotism, family)
4. e_4: **Authority / Subversion** (leadership, respect for social order, tradition)
5. e_5: **Sanctity / Degradation** (purity, religious/cultural sanctity, disgust avoidance)
6. e_6: **Liberty / Oppression** (resistance to domination and tyranny)

Each ecosystem consists of a two-edge loop (a media diet followed by a social circle). The progressive ecosystem shifts Authority into Care and Sanctity into Fairness. The conservative ecosystem shifts Care into Loyalty, Fairness into Sanctity, and half-strength Fairness into Authority:

If $R_{i\to j}(\theta)$ is the identity matrix except for entries $(i,i) = (j,j) = \cos\theta$, $(i,j) = -\sin\theta$, and $(j,i) = \sin\theta$. It moves weight from axis i toward axis j, then the progressive and conservative edge functions are

$$P(\theta) = R_{4\to1}(\theta)\, R_{5\to2}(\theta),$$

$$C(\theta) = R_{1\to3}(\theta)\, R_{2\to5}(\theta)\, R_{2\to4}(\theta/2)$$

Holonomies:

$$H_{\text{Left}} = P(8°)\, P(10°) = \begin{pmatrix} 0.951 & 0 & 0 & 0.309 & 0 & 0 \\ 0 & 0.951 & 0 & 0 & 0.309 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -0.309 & 0 & 0 & 0.951 & 0 & 0 \\ 0 & -0.309 & 0 & 0 & 0.951 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$H_{\text{Right}} = C(8°)\, C(10°) = \begin{pmatrix} 0.951 & 0 & -0.309 & 0 & 0 & 0 \\ 0 & 0.939 & -0.151 & -0.309 & 0 & 0 \\ 0.309 & 0 & 0.951 & 0 & 0 & 0 \\ 0 & 0.155 & 0 & 0.988 & -0.012 & 0 \\ 0 & 0.307 & 0 & -0.037 & 0.951 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Both are genuine rotations. The Liberty axis (6) remains untouched by every plane used, keeping its value at 0.344 universally.

The Initial State ($t = 0$: The Pluralistic Citizen)
In a healthy, cross-pollinated civil society, an average citizen (or a moderate baseline) maintains a balanced valuation across all six foundations. They care about fairness and care but also respect institutional authority and communal loyalty without any single axis dominating their cognitive geometry.

The shared baseline of the pluralistic citizen, normalized to a unit vector, is:

$$v_0 = 1/1.0161 \begin{pmatrix} 0.50 \\ 0.50 \\ 0.40 \\ 0.40 \\ 0.30 \\ 0.35 \end{pmatrix} \sim \begin{pmatrix} 0.492 \\ 0.492 \\ 0.394 \\ 0.394 \\ 0.295 \\ 0.344 \end{pmatrix}$$

Constructing Partisan Transition Matrices (U_{Left} and U_{Right})
Political ecosystems act as structured context loops (e.g., media diets, partisan universities, hyper-targeted social media algorithms). These environments apply systematic rotations in $SO(6)$ that amplify certain foundational axes while suppressing others.

Because rotations in $SO(6)$ operate across 2D planes ($\frac{6\times5}{2} = 15$ independent rotational planes), partisan media environments heavily weight rotations involving the **Care/Fairness vs. Authority/Sanctity** axes.

The Progressive Media Ecosystem (U_{Left})
A progressive institutional loop rotates the agent's frame to heavily elevate Care (e_1) and Fairness (e_2), while downplaying or treating Authority (e_4) and Sanctity (e_5) as irrelevant or harmful.

Outbound Transition Matrix (U_{Left}): Built via block rotations in the e_1-e_4 and e_2-e_5 planes, tilting the agent strongly toward universalist care and egalitarian fairness.

The Conservative Media Ecosystem (U_{Right})
A conservative institutional loop rotates the agent's frame to elevate group-binding foundations—balancing Care and Fairness with high valuations on Loyalty (e_3), Authority (e_4), and Sanctity (e_5).

Outbound Transition Matrix (U_{Right}): Built via block rotations that tilt the vector toward traditional order, in-group loyalty, and moral purity.

The Final Asymptotic States ($t \to \infty$)
After repeated traversals of their respective partisan loops without cross-group mixing, the agents converge onto their party's **eigenvector attractor**:

Target for Left Agent (Progressive Attractor v^*_{Left}) is dominated entirely by e_1Care and e_2 Fairness; e_4 Authority and e_5 Sanctity approach zero

$$v^*_{\text{Left}} = \begin{pmatrix} 0.65 \\ 0.65 \\ 0.15 \\ 0.10 \\ 0.05 \\ 0.35 \end{pmatrix}$$

Target for Right Agent (Conservative Attractor v^*_{Right}) is balanced across e_1 Care and e_2 Fairness, but sharply spiked on e_3 Loyalty,e_4 Authority, and e_5 Sanctity

$$v^*_{\text{Right}} = \begin{pmatrix} 0.35 \\ 0.30 \\ 0.55 \\ 0.50 \\ 0.45 \\ 0.25 \end{pmatrix}$$

Calculating the Holonomies and Incommensurability
The loop holonomy $H = U_{\text{back}} \cdot U_{\text{out}}$ aggregates the agent's political journey. Because H_{Left} and H_{Right} pull the agents into orthogonal regions of $\mathbb{R}^6$, their final gauge-invariant distance hits a maximum polarization ceiling:

$$\lim_{t\to\infty} d(\text{Left,Right}) = \delta_{\max}$$

The Mathematical Explanation for "Moral Muteness":
When Agent Left evaluates a conservative argument about national borders or traditional institutions (which relies heavily on e_3 Loyalty and e_5 Sanctity), Agent Left's internal coordinate frame projects that argument through their own axes (e_1 Care and e_2 Fairness).

Because their frame has rotated away from the binding foundations, **the conservative argument literally translates inside their geometry as a violation of Care/Harm**. This explains Haidt's central finding: partisans do not just disagree - their transformed coordinate systems render opposing moral languages mathematically uninterpretable, locking them into permanent divergence unless disrupted by the share ρ of encounters with strangers and shared institutional forcing.

The simple model fits to Haidt's central findings. Using a few improvements, it can be modified to fit the data from Jonathan Haidt well. He has collected experimental data using a "Moral Foundation Questionnaire".